

```
In [1]: !pip install fg-data-profiling pandas numpy --quiet
```

```
In [2]: import pandas as pd
import numpy as np
import data_profiling
from data_profiling import ProfileReport
```

```
In [3]: #Loading all datasets to be reviewed and cleaned
df1 = pd.read_csv('/home/jburkes/My_Projects/Operational_Efficiency/csv/raw/Cityworks WOs - 2020.csv')
df2 = pd.read_csv('/home/jburkes/My_Projects/Operational_Efficiency/csv/raw/Cityworks WOs - 2021.csv')
df3 = pd.read_csv('/home/jburkes/My_Projects/Operational_Efficiency/csv/raw/Cityworks WOs - 2022.csv')
df4 = pd.read_csv('/home/jburkes/My_Projects/Operational_Efficiency/csv/raw/Cityworks WOs - 2023.csv')
df5 = pd.read_csv('/home/jburkes/My_Projects/Operational_Efficiency/csv/raw/Cityworks WOs - 2024.csv')
df6 = pd.read_csv('/home/jburkes/My_Projects/Operational_Efficiency/csv/raw/Cityworks WOs - 2025.csv')
```

```
In [82]: #Concatenating all datasets
df = pd.concat([df1, df2, df3, df4, df5, df6], axis=0, ignore_index=True)

# Displaying merged dataset
df.head()
```

Out[82]:

	WorkOrderId	Description	Priority	Status	Submit To	Proj Start Date	Proj Finish Date	Actual Start	Actual Finish	Shop	...	Total Cost	Labor Cost	Comments
0	354966	TRIM BY IN- HOUSE	2	CLOSED	UF, IN-HOUSE	2020-01-02 08:47 AM	2020-01-12 08:47 AM	NaN	2020-01-30 12:00 PM	K	...	115.0	115.0	printed
1	354967	TRIM BY CONTRACTOR	3	CLOSED	UF, CONTRACTOR	2020-01-02 09:20 AM	2020-02-01 09:20 AM	2020-01-07 12:00 PM	2020-03-19 12:00 PM	I	...	136.0	136.0	completed - invoice # 100138.
2	354968	TRIM BY CONTRACTOR	3	CLOSED	UF, CONTRACTOR	2020-01-02 10:15 AM	2020-02-01 10:15 AM	2020-01-07 12:00 PM	2020-01-13 12:00 PM	D	...	105.0	105.0	inv10149
3	354969	TRIM BY IN- HOUSE	2	CLOSED	UF, IN-HOUSE	2020-01-03 08:32 AM	2020-01-13 08:32 AM	NaN	2020-01-09 12:00 PM	H	...	136.0	136.0	printed
4	354970	TRIM BY CONTRACTOR	3	CLOSED	UF, CONTRACTOR	2020-01-06 08:26 AM	2020-02-05 08:26 AM	2020-01-07 12:00 PM	2020-01-10 12:00 PM	I	...	115.0	115.0	inv 10148

5 rows × 26 columns

```
In [5]: #Generating an initial Profile Report of the merged dataset to obtain a baseline understanding of the st
profile = ProfileReport(df, title="Data Profile Report", explorative=True)
profile.to_notebook_iframe()
```

Summarize dataset: 0%| | 0/5 [00:00<?, ?it/s]

```

0%|
| 0/26 [00:00<?, ?it/s]
4%|██████
| 1/26 [00:00<00:04, 5.12it/s]
12%|██████████
| 3/26 [00:00<00:02, 9.38it/s]
15%|██████████
| 4/26 [00:00<00:02, 7.92it/s]
19%|██████████
| 5/26 [00:01<00:04, 4.40it/s]
23%|██████████
| 6/26 [00:22<02:21, 7.07s/it]
27%|██████████
| 7/26 [00:22<01:33, 4.91s/it]
31%|██████████
| 8/26 [00:23<01:06, 3.72s/it]
35%|██████████
| 9/26 [00:28<01:10, 4.15s/it]
50%|██████████
| 13/26 [00:29<00:20, 1.55s/it]
54%|██████████
| 14/26 [00:29<00:15, 1.30s/it]
100%|██████████
██████████| 26/26 [00:41<00:00, 1.61s/it]
Generate report structure: 0%| 0/1 [00:00<?, ?it/s]
Render HTML: 0%| 0/1 [00:00<?, ?it/s]

```



Data Profile Report

Overview

Overview

Alerts 35

Reproduction

Dataset statistics

Number of variables	26
Number of observations	29160
Missing cells	36995
Missing cells (%)	4.9%
Duplicate rows	0
Duplicate rows (%)	0.0%
Total size in memory	40.3 MiB
Average record size in memory	1.4 KiB

Variable types

Numeric	5
Categorical	11
DateTime	5
Text	5

Variables

Select Columns

WorkOrderId

```
In [83]: #Standardizing column headings to remove spaces and make all lower case  
df.columns = (df.columns.str.replace(' ', '_').str.lower().str.strip())  
df
```

Out[83]:

	workorderid	description	priority	status	submit_to	proj_start_date	proj_finish_date	actual_start	actual_finish	sf
0	354966	TRIM BY IN-HOUSE	2	CLOSED	UF, IN-HOUSE	2020-01-02 08:47 AM	2020-01-12 08:47 AM	NaN	2020-01-30 12:00 PM	
1	354967	TRIM BY CONTRACTOR	3	CLOSED	UF, CONTRACTOR	2020-01-02 09:20 AM	2020-02-01 09:20 AM	2020-01-07 12:00 PM	2020-03-19 12:00 PM	
2	354968	TRIM BY CONTRACTOR	3	CLOSED	UF, CONTRACTOR	2020-01-02 10:15 AM	2020-02-01 10:15 AM	2020-01-07 12:00 PM	2020-01-13 12:00 PM	
3	354969	TRIM BY IN-HOUSE	2	CLOSED	UF, IN-HOUSE	2020-01-03 08:32 AM	2020-01-13 08:32 AM	NaN	2020-01-09 12:00 PM	
4	354970	TRIM BY CONTRACTOR	3	CLOSED	UF, CONTRACTOR	2020-01-06 08:26 AM	2020-02-05 08:26 AM	2020-01-07 12:00 PM	2020-01-10 12:00 PM	
...	...	...	...	...	...	...	...	...	...	...
29155	404649	REMOVAL & STUMP GRINDING BY CONTRACTOR	3	ISSUED	UF, CONTRACTOR	2025-12-31 12:28 PM	2026-01-30 12:28 PM	2026-06-18 12:00 PM	NaN	
29156	404650	REMOVAL & STUMP GRINDING BY CONTRACTOR	3	CLOSED	UF, CONTRACTOR	2025-12-31 01:02 PM	2026-01-30 01:02 PM	2025-12-31 12:00 PM	2026-03-12 12:00 PM	
29157	404651	EMERGENCY REMOVAL BY IN-HOUSE	1	CLOSED	UF, IN-HOUSE	2025-12-31 01:07 PM	2026-01-01 01:07 PM	NaN	2026-01-08 12:24 PM	
29158	404652	REMOVAL & STUMP GRINDING BY CONTRACTOR	3	INPROG	UF, CONTRACTOR	2025-12-31 01:45 PM	2026-01-30 01:45 PM	NaN	NaN	

	workorderid	description	priority	status	submit_to	proj_start_date	proj_finish_date	actual_start	actual_finish	sf
29159	407842	TREE/SEEDLING PLANTING	4	CLOSED	UF, IN-HOUSE	2025-11-21 09:03 AM	2026-11-21 09:03 AM	NaN	2026-11-21 12:00 PM	N

29160 rows × 26 columns

```
In [84]: #Calculating the sum of all missing values for each column
df.isna().sum()
```

```
Out[84]: workorderid          0
description          0
priority             0
status               0
submit_to            7
proj_start_date      2
proj_finish_date     2
actual_start        12597
actual_finish        1823
shop                 376
map_page             19
address              3
location             67
category             65
date_initiated       0
project             12094
total_cost           0
labor_cost           0
comments            1202
zone                 99
units_accomplished   250
request_ids          1712
resolution           1539
sr_problem_code      1712
sr_description        1712
sr_priority          1714
dtype: int64
```

```
In [85]: #Displaying dataframe to review the rows with null values in proj_start_date
df[df['proj_start_date'].isnull()]
```

```
Out[85]:
```

	workorderid	description	priority	status	submit_to	proj_start_date	proj_finish_date	actual_start	actual_finish	sho
--	-------------	-------------	----------	--------	-----------	-----------------	------------------	--------------	---------------	-----

19836	381210	FMD Emergency Removal	3	CLOSED	NaN	NaN	NaN	2024-07-01 06:30 AM	2024-07-01 10:30 AM	t
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28461	402911	REMOVAL & STUMP GRINDING BY CONTRACTOR	3	CLOSED	UF, CONTRACTOR	NaN	NaN	2026-02-13 12:00 PM	2026-05-14 12:00 PM	
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2 rows × 26 columns

```
In [86]: #Replacing null values in proj_start_date with values from actual_start
df['proj_start_date'] = df['proj_start_date'].fillna(df['actual_start'])

#Replacing null values in proj_finish_date with values from actual_finish
df['proj_finish_date'] = df['proj_finish_date'].fillna(df['actual_finish'])
```

```
In [87]: #Displaying dataframe to review the rows with null values in submit_to
df[df['submit_to'].isnull()]
```

Out[87]:

	workorderid	description	priority	status	submit_to	proj_start_date	proj_finish_date	actual_start	actual_finish	shop
19836	381210	FMD Emergency Removal	3	CLOSED	NaN	2024-07-01 06:30 AM	2024-07-01 10:30 AM	2024-07-01 06:30 AM	2024-07-01 10:30 AM	H
25788	396606	FMD Emergency Removal	1	INPROG	NaN	2025-05-09 09:33 AM	2025-05-09 09:33 AM	NaN	NaN	NaN
25789	396607	FMD Emergency Removal	1	INPROG	NaN	2025-05-09 09:44 AM	2025-05-09 09:44 AM	NaN	NaN	C
27186	399637	TREE/SEEDLING PLANTING	4	HOLD	NaN	2025-07-25 03:42 PM	2026-07-25 03:42 PM	NaN	NaN	C
27966	401649	EMERGENCY DEBRIS PICK-UP	1	CLOSED	NaN	2025-09-22 11:30 AM	2025-09-23 11:30 AM	NaN	2025-11-03 02:02 PM	NaN
28720	403486	TREE/SEEDLING PLANTING	4	HOLD	NaN	2025-11-20 01:45 PM	2026-11-20 01:45 PM	NaN	NaN	C
28981	404195	TREE/SEEDLING PLANTING	4	HOLD	NaN	2025-12-15 10:03 AM	2026-12-15 10:03 AM	NaN	NaN	C

7 rows × 26 columns

In [88]: *#Replacing null values in submit_to with known values*

```
df['submit_to'] = df['submit_to'].fillna("UF, IN-HOUSE")
```

```
In [89]: #Replacing null values in actual_start with values from proj_start_date
df['actual_start'] = df['actual_start'].fillna(df['proj_start_date'])

#Replacing null values in actual_finish with values from proj_finish_date
df['actual_finish'] = df['actual_finish'].fillna(df['proj_finish_date'])
```

```
In [90]: #Calculating the sum of all missing values for each column
df.isna().sum()
```

```
Out[90]: workorderid          0
description          0
priority             0
status              0
submit_to           0
proj_start_date      0
proj_finish_date     0
actual_start         0
actual_finish        0
shop                376
map_page            19
address              3
location            67
category            65
date_initiated       0
project            12094
total_cost           0
labor_cost           0
comments            1202
zone                99
units_accomplished   250
request_ids          1712
resolution           1539
sr_problem_code      1712
sr_description       1712
sr_priority          1714
dtype: int64
```

```
In [91]: #Displaying 'description', 'sr_description' and 'category' columns to help determine value for 'category'
df_category_nulls = df[['description', 'sr_description', 'category']]
```

```
df_category_nulls[df_category_nulls['category'].isnull()]
```

Out[91]:

	description	sr_description	category
50	INSPECTION	NaN	NaN
51	INSPECTION	NaN	NaN
143	INSPECTION	NaN	NaN
164	INSPECTION	NaN	NaN
262	INSPECTION	NaN	NaN
...	...	...	...
4777	INSPECTION	NaN	NaN
4960	INSPECTION	NaN	NaN
4961	INSPECTION	NaN	NaN
4982	INSPECTION	NaN	NaN
5091	INSPECTION	NaN	NaN

65 rows × 3 columns

```
In [92]: #Dropping all rows where 'category' is null (NaN). Data owner reports these are phantom/duplicate record
df = df.dropna(subset=['category'])
```

```
In [93]: #Dropping all rows where 'address' is null (NaN). Data owner reports these are phantom/duplicate records
df = df.dropna(subset=['address'])
```

```
In [94]: #Resetting the index back to a clean sequence
df = df.reset_index(drop=True)
```

```
In [95]: #Calculating the sum of all missing values for each column
df.isna().sum()
```

```
Out[95]: workorderid      0
         description      0
         priority        0
         status          0
         submit_to       0
         proj_start_date  0
         proj_finish_date 0
         actual_start     0
         actual_finish    0
         shop            360
         map_page        3
         address         0
         location        1
         category        0
         date_initiated   0
         project        12028
         total_cost       0
         labor_cost       0
         comments        1140
         zone            32
         units_accomplished 248
         request_ids      1646
         resolution      1536
         sr_problem_code   1646
         sr_description    1646
         sr_priority      1647
         dtype: int64
```

```
In [96]: #Displaying 'address', 'zone' and 'map_page' columns to help determine value for 'map_page' nulls
df_map_page_nulls = df[['address', 'zone', 'map_page']]
df_map_page_nulls[df_map_page_nulls['map_page'].isnull()]
```

```
Out[96]:
```

	address	zone	map_page
82	600 MEMORIAL,HOUSTON TX	NaN	NaN
10574	7600 WILCREST, HOUSTON, TX	NaN	NaN
25723	Lawrence park Houston tx	NaN	NaN

```
In [97]: #Displaying dataframe info to see row index values
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 29092 entries, 0 to 29091
Data columns (total 26 columns):
#   Column                Non-Null Count  Dtype
---  -
0   workorderid           29092 non-null  int64
1   description           29092 non-null  object
2   priority              29092 non-null  int64
3   status                29092 non-null  object
4   submit_to             29092 non-null  object
5   proj_start_date       29092 non-null  object
6   proj_finish_date      29092 non-null  object
7   actual_start          29092 non-null  object
8   actual_finish         29092 non-null  object
9   shop                  28732 non-null  object
10  map_page              29089 non-null  object
11  address               29092 non-null  object
12  location              29091 non-null  object
13  category              29092 non-null  object
14  date_initiated        29092 non-null  object
15  project               17064 non-null  object
16  total_cost            29092 non-null  float64
17  labor_cost            29092 non-null  float64
18  comments              27952 non-null  object
19  zone                  29060 non-null  object
20  units_accomplished    28844 non-null  float64
21  request_ids           27446 non-null  object
22  resolution            27556 non-null  object
23  sr_problem_code       27446 non-null  object
24  sr_description        27446 non-null  object
25  sr_priority           27445 non-null  float64
dtypes: float64(4), int64(2), object(20)
memory usage: 5.8+ MB
```

```
In [98]: #Replacing null values in 'map_page' with known values
df.iloc[82, 10] = '492B'
df.iloc[10574, 10] = '533A'
df.iloc[25723, 10] = '454D'
```

```
In [99]: #Displaying dataframe to review the rows with null values in 'zone'  
df_zone_nulls = df[['address', 'map_page', 'zone']]  
df_zone_nulls[df_zone_nulls['zone'].isnull()]
```

Out[99]:

	address	map_page	zone
82	600 MEMORIAL,HOUSTON TX	492B	NaN
2888	9455 W MONTGOMERY RD, HOUSTON, TX, 77088	412N	NaN
2901	8514 W MONTGOMERY RD, HOUSTON, TX, 77088	412S	NaN
2941	1466 WILLOW ROCK RD, HOUSTON, TX, 77088	412P	NaN
2943	1426 WILLOW ROCK RD, HOUSTON, TX, 77088	412Q	NaN
2981	1606 RAINY RIVER DR, HOUSTON, TX, 77088	412K	NaN
3001	1623 TWINBROOKE DR, HOUSTON, TX, 77088	412K	NaN
7635	13500 BONILLA LN, TX, 77083	528K	NaN
8041	12834 COVEY LN, HOUSTON, TX, 77099	528V	NaN
8507	5807 MID PINES ST	457S	NaN
10574	7600 WILCREST, HOUSTON, TX	533A	NaN
12389	Lake Houston Wilderness Park	257Q	NaN
12827	25840 FM-1485 E, New Caney, Texas, 77357 (L...	257Q	NaN
14364	25840 FM-1485 E, New Caney, Texas, 77357 (Lake...	257Q	NaN
15569	25840 FM-1485 E, New Caney, Texas, 77357 (Lake...	257Q	NaN
18275	Lake Houston Wilderness Park	257U	NaN
18881	2721 Mierianne St, Houston, Texas, 77093	414N	NaN
18884	25840 FM 1485 Rd (Lake Houston Wilderness Park)	257U	NaN
20192	6401 Kirby Dr, Houston, Texas, 77005	532C	NaN
20857	5155 Grape St, Houston, Texas, 77096	531Q	NaN
23806	2727 Wroxton Rd, Houston, Texas, 77005	532C	NaN
23808	2920 Wroxton Rd, Houston, Texas, 77005	492X	NaN
24160	12300 Aldine Westfield Rd, Houston, Texas, 77093	413M	NaN
24382	5503 Briarwood Forest Dr, Houston, Texas, 77088	411L	NaN

	address	map_page	zone
24594	11703 Davenwood Cir, Houston, Texas, 77089	616D	NaN
24905	12300 Aldine Westfield Rd, Houston, Texas, ...	413M	NaN
24988	10506 Sagevale Ln, Houston, Texas, 77089	576W	NaN
24989	5015 Aldine Bender Rd, Houston, Texas, 77032	374U	NaN
24990	9502 Sageknoll Ln, Houston, Texas, 77089	576W	NaN
25723	Lawrence park Houston tx	454D	NaN
26340	15634 Sunset Rock Dr, Houston, Texas, 77084	448B	NaN
27573	11410 Ryland Rd, Houston, Texas, 77066 (Haz...	371Q	NaN

In [100... *#Replacing null values in 'zone' with known values*

```
df.iloc[82, 19] = '3, ZONE'
df.iloc[2888, 19] = '3, ZONE'
df.iloc[2901, 19] = '3, ZONE'
df.iloc[2941, 19] = '3, ZONE'
df.iloc[2943, 19] = '3, ZONE'
df.iloc[2981, 19] = '3, ZONE'
df.iloc[3001, 19] = '3, ZONE'
df.iloc[7635, 19] = '2, ZONE'
df.iloc[8041, 19] = '2, ZONE'
df.iloc[8507, 19] = '4, ZONE'
df.iloc[10574, 19] = '5, ZONE'
df.iloc[12389, 19] = '4, ZONE'
df.iloc[12827, 19] = '4, ZONE'
df.iloc[14364, 19] = '4, ZONE'
df.iloc[15569, 19] = '4, ZONE'
df.iloc[18275, 19] = '4, ZONE'
df.iloc[18881, 19] = '4, ZONE'
df.iloc[18884, 19] = '4, ZONE'
df.iloc[20192, 19] = '7, ZONE'
df.iloc[20857, 19] = '7, ZONE'
df.iloc[23806, 19] = '7, ZONE'
df.iloc[23808, 19] = '7, ZONE'
df.iloc[24160, 19] = '4, ZONE'
df.iloc[24382, 19] = '3, ZONE'
```

```
df.iloc[24594, 19] = '6, ZONE'
df.iloc[24905, 19] = '4, ZONE'
df.iloc[24988, 19] = '5, ZONE'
df.iloc[24989, 19] = '4, ZONE'
df.iloc[24990, 19] = '5, ZONE'
df.iloc[25723, 19] = '4, ZONE'
df.iloc[26340, 19] = '3, ZONE'
df.iloc[27573, 19] = '4, ZONE'
```

```
In [101... #Re-calculating the sum of all missing values for each column
df.isna().sum()
```

```
Out[101... workorderid          0
description          0
priority             0
status              0
submit_to           0
proj_start_date      0
proj_finish_date     0
actual_start         0
actual_finish        0
shop                 360
map_page             0
address              0
location             1
category             0
date_initiated       0
project             12028
total_cost           0
labor_cost           0
comments            1140
zone                 0
units_accomplished   248
request_ids          1646
resolution           1536
sr_problem_code      1646
sr_description       1646
sr_priority          1647
dtype: int64
```

```
In [102... #Replacing null values in shop with known values by zone
df.loc[(df['shop'].isnull()) & (df['zone'] == "2, ZONE"), 'shop'] = 'G'
```

```
df.loc[(df['shop'].isnull() & (df['zone'] == "3, ZONE"), 'shop'] = 'A'
df.loc[(df['shop'].isnull() & (df['zone'] == "4, ZONE"), 'shop'] = 'H'
df.loc[(df['shop'].isnull() & (df['zone'] == "5, ZONE"), 'shop'] = 'E'
df.loc[(df['shop'].isnull() & (df['zone'] == "6, ZONE"), 'shop'] = 'D'
df.loc[(df['shop'].isnull() & (df['zone'] == "7, ZONE"), 'shop'] = 'K'
```

```
In [103... #Re-calculating the sum of all missing values for each column
df.isna().sum()
```

```
Out[103... workorderid          0
description          0
priority             0
status              0
submit_to           0
proj_start_date      0
proj_finish_date     0
actual_start         0
actual_finish        0
shop                 0
map_page             0
address              0
location             1
category             0
date_initiated       0
project              12028
total_cost           0
labor_cost           0
comments             1140
zone                 0
units_accomplished   248
request_ids          1646
resolution           1536
sr_problem_code       1646
sr_description        1646
sr_priority           1647
dtype: int64
```

```
In [104... #Adding a new column named 'contractor'
df['contractor'] = None
df
```

Out[104...

	workorderid	description	priority	status	submit_to	proj_start_date	proj_finish_date	actual_start	actual_finish	sf
0	354966	TRIM BY IN-HOUSE	2	CLOSED	UF, IN-HOUSE	2020-01-02 08:47 AM	2020-01-12 08:47 AM	2020-01-02 08:47 AM	2020-01-30 12:00 PM	
1	354967	TRIM BY CONTRACTOR	3	CLOSED	UF, CONTRACTOR	2020-01-02 09:20 AM	2020-02-01 09:20 AM	2020-01-07 12:00 PM	2020-03-19 12:00 PM	
2	354968	TRIM BY CONTRACTOR	3	CLOSED	UF, CONTRACTOR	2020-01-02 10:15 AM	2020-02-01 10:15 AM	2020-01-07 12:00 PM	2020-01-13 12:00 PM	
3	354969	TRIM BY IN-HOUSE	2	CLOSED	UF, IN-HOUSE	2020-01-03 08:32 AM	2020-01-13 08:32 AM	2020-01-03 08:32 AM	2020-01-09 12:00 PM	
4	354970	TRIM BY CONTRACTOR	3	CLOSED	UF, CONTRACTOR	2020-01-06 08:26 AM	2020-02-05 08:26 AM	2020-01-07 12:00 PM	2020-01-10 12:00 PM	
...	...	...	...	...	...	...	...	...	...	
29087	404649	REMOVAL & STUMP GRINDING BY CONTRACTOR	3	ISSUED	UF, CONTRACTOR	2025-12-31 12:28 PM	2026-01-30 12:28 PM	2026-06-18 12:00 PM	2026-01-30 12:28 PM	
29088	404650	REMOVAL & STUMP GRINDING BY CONTRACTOR	3	CLOSED	UF, CONTRACTOR	2025-12-31 01:02 PM	2026-01-30 01:02 PM	2025-12-31 12:00 PM	2026-03-12 12:00 PM	
29089	404651	EMERGENCY REMOVAL BY IN-HOUSE	1	CLOSED	UF, IN-HOUSE	2025-12-31 01:07 PM	2026-01-01 01:07 PM	2025-12-31 01:07 PM	2026-01-08 12:24 PM	
29090	404652	REMOVAL & STUMP GRINDING BY CONTRACTOR	3	INPROG	UF, CONTRACTOR	2025-12-31 01:45 PM	2026-01-30 01:45 PM	2025-12-31 01:45 PM	2026-01-30 01:45 PM	

	workorderid	description	priority	status	submit_to	proj_start_date	proj_finish_date	actual_start	actual_finish	sl
29091	407842	TREE/SEEDLING PLANTING	4	CLOSED	UF, IN-HOUSE	2025-11-21 09:03 AM	2026-11-21 09:03 AM	2025-11-21 09:03 AM	2026-11-21 12:00 PM	

29092 rows × 27 columns

```
In [105... #Replacing values in 'contractor' with known tree maintenance contractors by zone
df.loc[(df['contractor'].isnull()) & (df['zone'] == "1, ZONE") & (df['submit_to'] == "UF, CONTRACTOR"),
df.loc[(df['contractor'].isnull()) & (df['zone'] == "2, ZONE") & (df['submit_to'] == "UF, CONTRACTOR"),
df.loc[(df['contractor'].isnull()) & (df['zone'] == "3, ZONE") & (df['submit_to'] == "UF, CONTRACTOR"),
df.loc[(df['contractor'].isnull()) & (df['zone'] == "4, ZONE") & (df['submit_to'] == "UF, CONTRACTOR"),
df.loc[(df['contractor'].isnull()) & (df['zone'] == "5, ZONE") & (df['submit_to'] == "UF, CONTRACTOR"),
df.loc[(df['contractor'].isnull()) & (df['zone'] == "6, ZONE") & (df['submit_to'] == "UF, CONTRACTOR"),
df.loc[(df['contractor'].isnull()) & (df['zone'] == "7, ZONE") & (df['submit_to'] == "UF, CONTRACTOR"),
```

```
In [107... #Replacing values in 'contractor' with known tree planting contractors by zone
df.loc[(df['description'] == "TREE/SEEDLING PLANTING") & (df['zone'] == "1, ZONE") & (df['submit_to'] ==
df.loc[(df['description'] == "TREE/SEEDLING PLANTING") & (df['zone'] == "2, ZONE") & (df['submit_to'] ==
df.loc[(df['description'] == "TREE/SEEDLING PLANTING") & (df['zone'] == "3, ZONE") & (df['submit_to'] ==
df.loc[(df['description'] == "TREE/SEEDLING PLANTING") & (df['zone'] == "4, ZONE") & (df['submit_to'] ==
df.loc[(df['description'] == "TREE/SEEDLING PLANTING") & (df['zone'] == "5, ZONE") & (df['submit_to'] ==
df.loc[(df['description'] == "TREE/SEEDLING PLANTING") & (df['zone'] == "6, ZONE") & (df['submit_to'] ==
df.loc[(df['description'] == "TREE/SEEDLING PLANTING") & (df['zone'] == "7, ZONE") & (df['submit_to'] ==
df
```

Out[107...

	workorderid	description	priority	status	submit_to	proj_start_date	proj_finish_date	actual_start	actual_finish	sf
0	354966	TRIM BY IN-HOUSE	2	CLOSED	UF, IN-HOUSE	2020-01-02 08:47 AM	2020-01-12 08:47 AM	2020-01-02 08:47 AM	2020-01-30 12:00 PM	
1	354967	TRIM BY CONTRACTOR	3	CLOSED	UF, CONTRACTOR	2020-01-02 09:20 AM	2020-02-01 09:20 AM	2020-01-07 12:00 PM	2020-03-19 12:00 PM	
2	354968	TRIM BY CONTRACTOR	3	CLOSED	UF, CONTRACTOR	2020-01-02 10:15 AM	2020-02-01 10:15 AM	2020-01-07 12:00 PM	2020-01-13 12:00 PM	
3	354969	TRIM BY IN-HOUSE	2	CLOSED	UF, IN-HOUSE	2020-01-03 08:32 AM	2020-01-13 08:32 AM	2020-01-03 08:32 AM	2020-01-09 12:00 PM	
4	354970	TRIM BY CONTRACTOR	3	CLOSED	UF, CONTRACTOR	2020-01-06 08:26 AM	2020-02-05 08:26 AM	2020-01-07 12:00 PM	2020-01-10 12:00 PM	
...	...	...	...	...	...	...	...	...	...	
29087	404649	REMOVAL & STUMP GRINDING BY CONTRACTOR	3	ISSUED	UF, CONTRACTOR	2025-12-31 12:28 PM	2026-01-30 12:28 PM	2026-06-18 12:00 PM	2026-01-30 12:28 PM	
29088	404650	REMOVAL & STUMP GRINDING BY CONTRACTOR	3	CLOSED	UF, CONTRACTOR	2025-12-31 01:02 PM	2026-01-30 01:02 PM	2025-12-31 12:00 PM	2026-03-12 12:00 PM	
29089	404651	EMERGENCY REMOVAL BY IN-HOUSE	1	CLOSED	UF, IN-HOUSE	2025-12-31 01:07 PM	2026-01-01 01:07 PM	2025-12-31 01:07 PM	2026-01-08 12:24 PM	
29090	404652	REMOVAL & STUMP GRINDING BY CONTRACTOR	3	INPROG	UF, CONTRACTOR	2025-12-31 01:45 PM	2026-01-30 01:45 PM	2025-12-31 01:45 PM	2026-01-30 01:45 PM	

	workorderid	description	priority	status	submit_to	proj_start_date	proj_finish_date	actual_start	actual_finish	sf
29091	407842	TREE/SEEDLING PLANTING	4	CLOSED	UF, IN-HOUSE	2025-11-21 09:03 AM	2026-11-21 09:03 AM	2025-11-21 09:03 AM	2026-11-21 12:00 PM	

29092 rows × 27 columns

```
In [110... #Replacing values in 'contractor' with known tree watering contractors by zone
df.loc[(df['description'] == "WATER TREES") & (df['zone'] == "1, ZONE") & (df['submit_to'] == "UF, CONTR
df.loc[(df['description'] == "WATER TREES") & (df['zone'] == "2, ZONE") & (df['submit_to'] == "UF, CONTR
df.loc[(df['description'] == "WATER TREES") & (df['zone'] == "3, ZONE") & (df['submit_to'] == "UF, CONTR
df.loc[(df['description'] == "WATER TREES") & (df['zone'] == "4, ZONE") & (df['submit_to'] == "UF, CONTR
df.loc[(df['description'] == "WATER TREES") & (df['zone'] == "5, ZONE") & (df['submit_to'] == "UF, CONTR
df.loc[(df['description'] == "WATER TREES") & (df['zone'] == "6, ZONE") & (df['submit_to'] == "UF, CONTR
df.loc[(df['description'] == "WATER TREES") & (df['zone'] == "7, ZONE") & (df['submit_to'] == "UF, CONTR
df
```

Out[110...

	workorderid	description	priority	status	submit_to	proj_start_date	proj_finish_date	actual_start	actual_finish	sf
0	354966	TRIM BY IN-HOUSE	2	CLOSED	UF, IN-HOUSE	2020-01-02 08:47 AM	2020-01-12 08:47 AM	2020-01-02 08:47 AM	2020-01-30 12:00 PM	
1	354967	TRIM BY CONTRACTOR	3	CLOSED	UF, CONTRACTOR	2020-01-02 09:20 AM	2020-02-01 09:20 AM	2020-01-07 12:00 PM	2020-03-19 12:00 PM	
2	354968	TRIM BY CONTRACTOR	3	CLOSED	UF, CONTRACTOR	2020-01-02 10:15 AM	2020-02-01 10:15 AM	2020-01-07 12:00 PM	2020-01-13 12:00 PM	
3	354969	TRIM BY IN-HOUSE	2	CLOSED	UF, IN-HOUSE	2020-01-03 08:32 AM	2020-01-13 08:32 AM	2020-01-03 08:32 AM	2020-01-09 12:00 PM	
4	354970	TRIM BY CONTRACTOR	3	CLOSED	UF, CONTRACTOR	2020-01-06 08:26 AM	2020-02-05 08:26 AM	2020-01-07 12:00 PM	2020-01-10 12:00 PM	
...	...	...	...	...	...	...	...	...	...	
29087	404649	REMOVAL & STUMP GRINDING BY CONTRACTOR	3	ISSUED	UF, CONTRACTOR	2025-12-31 12:28 PM	2026-01-30 12:28 PM	2026-06-18 12:00 PM	2026-01-30 12:28 PM	
29088	404650	REMOVAL & STUMP GRINDING BY CONTRACTOR	3	CLOSED	UF, CONTRACTOR	2025-12-31 01:02 PM	2026-01-30 01:02 PM	2025-12-31 12:00 PM	2026-03-12 12:00 PM	
29089	404651	EMERGENCY REMOVAL BY IN-HOUSE	1	CLOSED	UF, IN-HOUSE	2025-12-31 01:07 PM	2026-01-01 01:07 PM	2025-12-31 01:07 PM	2026-01-08 12:24 PM	
29090	404652	REMOVAL & STUMP GRINDING BY CONTRACTOR	3	INPROG	UF, CONTRACTOR	2025-12-31 01:45 PM	2026-01-30 01:45 PM	2025-12-31 01:45 PM	2026-01-30 01:45 PM	

	workorderid	description	priority	status	submit_to	proj_start_date	proj_finish_date	actual_start	actual_finish	sf
29091	407842	TREE/SEEDLING PLANTING	4	CLOSED	UF, IN-HOUSE	2025-11-21 09:03 AM	2026-11-21 09:03 AM	2025-11-21 09:03 AM	2026-11-21 12:00 PM	

29092 rows × 27 columns

```
In [111... #Saving cleaned data set as new csv file
df.to_csv('/home/jburkes/My_Projects/Operational_Efficiency/csv/cleaned/staging_cwwos.csv', index=False)
```

```
In [ ]:
```